

**COMPANION FOR CHAPTER 6**

**OF**

**FUNDAMENTALS  
OF PROBABILITY**  
WITH STOCHASTIC PROCESSES

**FOURTH EDITION**

**SAEED GHAHRAMANI**

Western New England University  
Springfield, Massachusetts, USA



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## Chapter 6

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# Continuous Random Variables

### 6A ADDITIONAL EXAMPLES

**Example 6a** Let  $X$  be a random variable (discrete or continuous) that satisfies  $0 \leq X \leq 1$  with probability 1. Show that  $\text{Var}(X) \leq 1/4$ .

*Solution:* Let  $\mu = E(X)$ . Since  $0 \leq X \leq 1$ , we have that  $X^2 \leq X$ , which yields  $E(X^2) \leq E(X)$ . Thus

$$\text{Var}(X) = E(X^2) - [E(X)]^2 \leq E(X) - [E(X)]^2 = \mu - \mu^2.$$

Now the function  $g(\mu) = \mu - \mu^2$  is maximum for a value of  $\mu$  for which  $g'(\mu) = 0$ , or  $1 - 2\mu = 0$ . Thus  $g$  is maximum at  $\mu = 1/2$ . The value of  $g$  at  $\mu = 1/2$  is  $g(1/2) = 1/2 - 1/4 = 1/4$ . So  $\mu - \mu^2 \leq 1/4$  for all values of  $\mu$ . Therefore,

$$\text{Var}(X) \leq \mu - \mu^2 \leq \frac{1}{4}. \quad \blacklozenge$$

**Example 6b** Let  $X$  be a continuous random variable with probability density function

$$f(x) = \begin{cases} c|x-2| & \text{if } -4 < x < 8 \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Calculate the value of  $c$ .
- (b) Find the distribution function of  $X$ .
- (c) Find  $E(X)$ .

*Solution:* (a) We must have

$$c \int_{-4}^2 (2-x) dx + c \int_2^8 (x-2) dx = 1.$$

This gives  $36c = 1$ , or  $c = 1/36$ .

**(b)** Note that

$$f(x) = \begin{cases} \frac{1}{36}(2-x) & -4 < x < 2 \\ \frac{1}{36}(x-2) & 2 \leq x < 8 \\ 0 & \text{otherwise.} \end{cases}$$

Let  $F$  be the distribution function of  $X$ . For  $-4 \leq t < 2$ ,

$$F(t) = \int_{-4}^t \frac{1}{36}(2-x) dx = -\frac{1}{72}t^2 + \frac{1}{18}t + \frac{4}{9},$$

and, for  $2 \leq t < 8$ ,

$$F(t) = \int_{-4}^2 \frac{1}{36}(2-x) dx + \int_2^t \frac{1}{36}(x-2) dx = \frac{1}{72}t^2 - \frac{1}{18}t + \frac{5}{9}.$$

Hence

$$F(t) = \begin{cases} 0 & t < -4 \\ -\frac{1}{72}t^2 + \frac{1}{18}t + \frac{4}{9} & -4 \leq t < 2 \\ \frac{1}{72}t^2 - \frac{1}{18}t + \frac{5}{9} & 2 \leq t < 8 \\ 1 & t \geq 8. \end{cases}$$

$$\text{(c)} \quad E(X) = \int_{-4}^2 x \cdot \frac{1}{36}(2-x) dx + \int_2^8 x \cdot \frac{1}{36}(x-2) dx = 2. \quad \blacklozenge$$

**Example 6c** Let  $X$  be a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{2}{3}(x-2) & 2 < x \leq 3 \\ \frac{1}{3}(5-x) & 3 < x < 5 \\ 0 & \text{otherwise.} \end{cases}$$

**(i)** Find  $F$ , the distribution function of  $X$ .

**(ii)** Calculate  $E(X)$ ,  $\text{Var}(X)$ , and  $E(|X-4|)$ .

*Solution:* **(i)** For  $2 \leq t < 3$ ,

$$F(t) = \int_2^t \frac{2}{3}(x-2) dx = \frac{1}{3}(t^2 - 4t + 4).$$

For  $3 \leq t < 5$ ,

$$\begin{aligned} F(t) &= \int_2^3 \frac{2}{3}(x-2) dx + \int_3^t \frac{1}{3}(5-x) dx \\ &= \frac{1}{3} + \left(-\frac{1}{6}t^2 + \frac{5}{3}t - \frac{7}{2}\right) = \frac{1}{6}(-t^2 + 10t - 19). \end{aligned}$$

Therefore,

$$F(t) = \begin{cases} 0 & t < 2 \\ \frac{1}{3}(t^2 - 4t + 4) & 2 \leq t < 3 \\ \frac{1}{6}(-t^2 + 10t - 19) & 3 \leq t < 5 \\ 1 & t \geq 5. \end{cases}$$

(ii) We have

$$\begin{aligned} E(X) &= \int_2^3 x \cdot \frac{2}{3}(x-2) dx + \int_3^5 x \cdot \frac{1}{3}(5-x) dx \\ &= \frac{8}{9} + \frac{22}{9} = \frac{10}{3} \approx 3.33. \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_2^3 x^2 \cdot \frac{2}{3}(x-2) dx + \int_3^5 x^2 \cdot \frac{1}{3}(5-x) dx \\ &= \frac{43}{18} + \frac{82}{9} = \frac{23}{2}. \end{aligned}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{23}{2} - \frac{100}{9} = \frac{7}{18} \approx 0.389.$$

$$\begin{aligned} E(|X-4|) &= \int_{-\infty}^{\infty} |x-4|f(x) dx \\ &= \int_2^3 (4-x) \cdot \frac{2}{3}(x-2) dx + \int_3^4 (4-x) \cdot \frac{1}{3}(5-x) dx + \\ &\quad \int_4^5 (x-4) \cdot \frac{1}{3}(5-x) dx = \frac{4}{9} + \frac{5}{18} + \frac{1}{18} = \frac{7}{9} \approx 0.778. \quad \blacklozenge \end{aligned}$$

**Example 6d** For  $\alpha > 0, \beta > 0$ , let

$$f(x) = \begin{cases} (e^{-\alpha x} - e^{-\beta x})/x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0. \end{cases}$$

Under what condition(s) is  $f$  a density function?

*Solution:*  $f(x) \geq 0$  implies that we must have  $e^{-\alpha x} > e^{-\beta x}$ , which requires  $\alpha < \beta$ . We must also have that

$$\int_0^\infty \frac{e^{-\alpha x} - e^{-\beta x}}{x} dx = 1. \quad (1)$$

To see under what conditions this is true, let

$$H(t) = \int_{0+}^\infty \frac{e^{-tx}(e^{-\alpha x} - e^{-\beta x})}{x} dx, \quad t \geq 0.$$

It is easily seen that  $H$  is differentiable and

$$\begin{aligned} H'(t) &= \int_0^\infty (e^{-\beta x} - e^{-\alpha x})e^{-tx} dx = \left[ \frac{e^{-(\alpha+t)x}}{\alpha+t} - \frac{e^{-(\beta+t)x}}{\beta+t} \right]_0^\infty \\ &= \frac{1}{\beta+t} - \frac{1}{\alpha+t}. \end{aligned}$$

Thus,

$$H(t) = \ln \frac{\beta+t}{\alpha+t} + c.$$

But since  $\lim_{t \rightarrow \infty} H(t) = 0$ , this relation gives  $0 = \ln 1 + c$  or  $c = 0$ , so

$$H(t) = \ln \frac{\beta+t}{\alpha+t}.$$

Now

$$\int_{0+}^\infty \frac{e^{-\alpha x} - e^{-\beta x}}{x} dx = H(0) = \ln \frac{\beta}{\alpha}.$$

Therefore, for the validity of (1), we must have  $\ln(\beta/\alpha) = 1$ , or  $\beta = e\alpha$ . If  $\beta = e\alpha$ , then  $\alpha < \beta$  is also true and  $f$  would be a density function. ♦

**Example 6e** A random variable  $X$  is said to be **logistic** if its probability density function is given by

$$f(x) = \frac{e^x}{(1+e^x)^2}, \quad -\infty < x < \infty.$$

- (a) Find the distribution function of a logistic random variable  $X$ .
- (b) Let  $Y$  be a random number between 0 and 1. Show that  $\ln \left( \frac{Y}{1-Y} \right)$  is logistic.

*Solution:* (a) Let  $F$  be the distribution function of  $X$ . For  $-\infty < t < \infty$ ,

$$F(t) = \int_{-\infty}^t \frac{e^x}{(1+e^x)^2} dx.$$

To calculate this integral, we make the substitution  $u = 1 + e^x$ . We have  $du = e^x dx$ . Therefore,

$$F(t) = \int_1^{1+e^t} \frac{1}{u^2} du = \left[ -\frac{1}{u} \right]_1^{1+e^t} = 1 - \frac{1}{1+e^t} = \frac{e^t}{1+e^t}, \quad -\infty < t < \infty.$$

(b) We will show that the distribution function of  $\ln \left( \frac{Y}{1-Y} \right)$  is  $\frac{e^t}{1+e^t}$ . By part (a), this shows that  $\ln \left( \frac{Y}{1-Y} \right)$  is logistic. We have

$$\begin{aligned} P\left(\ln \left( \frac{Y}{1-Y} \right) \leq t\right) &= P\left(\frac{Y}{1-Y} \leq e^t\right) = P(Y \leq (1-Y)e^t) \\ &= P\left(Y \leq \frac{e^t}{1+e^t}\right) = \frac{\frac{e^t}{1+e^t} - 0}{1-0} = \frac{e^t}{1+e^t}. \quad \blacklozenge \end{aligned}$$

**Example 6f** Let  $X$  be a continuous random variable with distribution function  $F$  and density function  $f$ . Find the distribution and the density functions of  $Y = |X|$ .

*Solution:* Since  $Y \geq 0$ , for  $t < 0$ ,  $P(Y \leq t) = 0$ . For  $t \geq 0$ ,

$$\begin{aligned} P(Y \leq t) &= P(|X| \leq t) = P(-t \leq X \leq t) = P(X \leq t) - P(X < -t) \\ &= P(X \leq t) - P(X \leq -t) = F(t) - F(-t). \end{aligned}$$

Hence  $G$ , the distribution function of  $|X|$  is given by

$$G(t) = \begin{cases} 0 & \text{if } t < 0 \\ F(t) - F(-t) & \text{if } t \geq 0 \end{cases}$$

and so  $g$ , the density function of  $|X|$  is obtained by differentiation of  $G$ :

$$g(t) = G'(t) = \begin{cases} f(t) + f(-t) & \text{if } t \geq 0 \\ 0 & \text{if } t < 0. \end{cases} \quad \blacklozenge$$

**Example 6g** Let  $X$  be a continuous random variable with the following density function.

$$f_X(x) = \begin{cases} 1/2 & \text{if } x \in (-1, 1) \\ 0 & \text{otherwise.} \end{cases}$$

Using the method of transformations, find the density function of  $Y = e^X$ .

*Solution:* The set of possible values of  $X$  is  $A = (-1, 1)$ . Let  $h : (-1, 1) \rightarrow \mathbf{R}$  be defined by  $h(x) = e^x$ . We want to find the density function of  $Y = h(X) = e^X$ . The set of possible values of  $h(X)$  is  $B = \{h(a) : a \in A\} = (1/e, e)$ . The function  $h$  is invertible

with the inverse  $x = g(y) = \ln y$  which is differentiable and its derivative is  $g'(y) = 1/y$ . Therefore, by Theorem 6.1 of the book,

$$f_Y(y) = f_X(g(y))|g'(y)| = \frac{1}{2} \left| \frac{1}{y} \right| = \frac{1}{2y}, \quad y \in (1/e, e),$$

is the density function of  $Y = e^X$ .  $\blacklozenge$